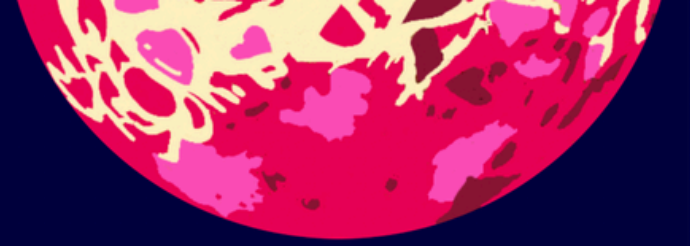
TOPICS



The material of reality

How is physical reality built

what are cells, how they live?

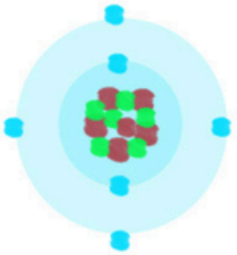
Our crucial sources - food, air, water.
How are they managed? & Why is there a need for unity on this

water and plastic issue: solution ideas and examples

in some matters improved and perfected World vision

Primary structure of our reality

Everything physical in our surrounding is made of **atoms.**
Atoms are made of 3 types of teeny tiny particles (balls alike structures?), that are arranged in a specific system. 2 of them can carry energy.



In the central part there are two particles called protons and neutrons stuck together into what we call a nucleus. Protons can carry positive electric charge*. Neutrons can't carry any electric charge.

electrons
protons
neutrons

Third and last on this scale (as far as we know) particles are electrons. They can carry negative charge. They move in specified areas around the nucleus. It is specified on what level are they (explanation on a graphic below), but not specifically where on that level are they. They are somewhere in what we call a probability cloud. In each level there can be a specified amount of electrons with specified amount of energy they carry.

singular carbon atom

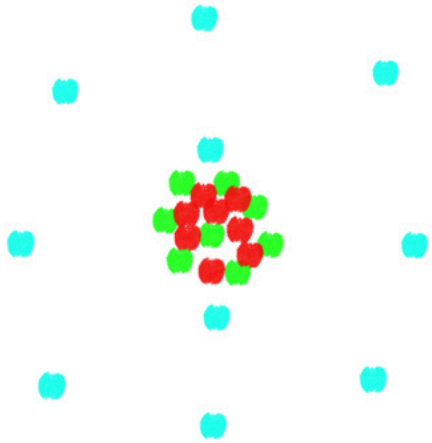
70 kg of human is approximately made
of
600 000 000 000 000 000 000 000 000
of
carbon atoms

*current state of knowledge can hardly explain why are theese different charges different, because separately they work the same. What we know for instance is that the same charges (+&+ / -&-) repel each other and opposite charges (+&-) attract.

Electrons associated with certain nucleus, in atoms' basic state, have **altogether** the same amount of energy as the nucleus, but with a different charge (negative instead of positive).
Atoms can connect ogether via electrons.

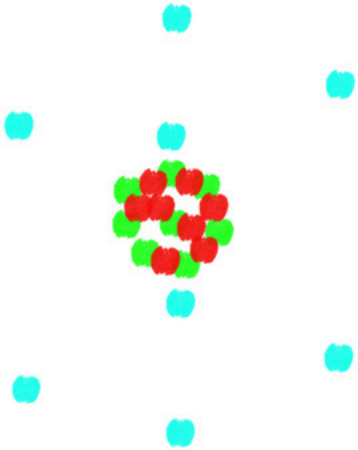
appearing in next topic

oxygen



8 protons, 8 neutrons and 8 electrons

oxygen with two extra electrons



hydrogen



1 proton, 1 neutron and 1 electron

proton (hydrogen with no electron)



we differentiate atoms into elements. Every element we know is described in Mendelev's periodic table. They differ in the amount of electrons (therefore protons and neutrons) they have. For example one oxygen has 8 electrons, one helium has 2 electrons, one carbon has 6 electrons

Atoms connect together by connecting the outer parts - electrons.
It precisely means connecting specified electrons fields. So after the atoms connect, they give one another the acces to their field. Now some electrons from last fields of connected atoms can move around electrons fields of two (or more) atoms.

Nevertheless no parts of atoms (connected or not) touch.

Nucleus of two atoms repel one another, because they have the same electric charge (+&+).
Electrons of the same and different atoms repel one another, because they have the same electric charge (-&-).
Aa single electron from each atom can't touch the nucleus, because it doesn't have enough energy on it's own to do so.

Facts, that will be useful further in the further parts of this
on Earth LIfe magazine

1.
Everything physical is made of atoms.

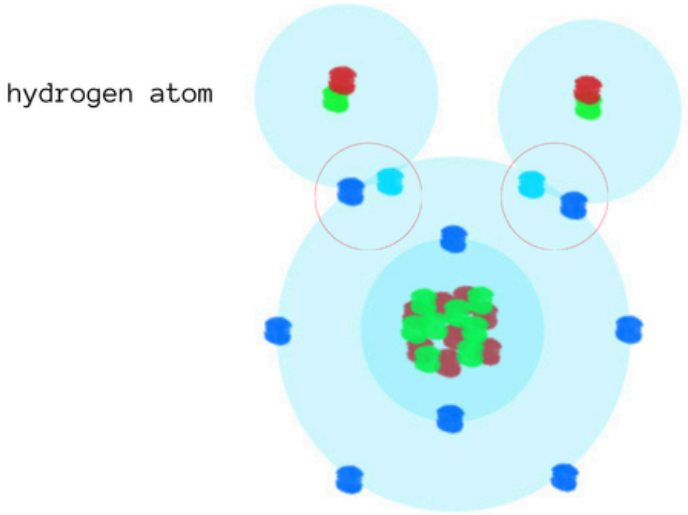
Everything that we call alive is made of cells.

Cells are of course made of atoms.

Cells' atoms are arranged in a way in which by their collective work they can obtain the energy from the environment and in some way *make it flow throughout* them.

Cells without the energy flow in them, are considered 'dead'.

singular water molecule



Oxygen electrons

Hydrogens electrons

Shared electrons

shared = some other atoms electron that now can move in another ones field

in one water (H20) molecule each hydrogen shares electron from its first field with one electron from the second electron field of oxygen. So in water molecule each hydrogen has 2 (1 of them shared) electrons instead of one now, and water has 10 instead of 8 (2 of them shared from & with the two hydrogens)

N E X T T O P I C

How your cells obtain and use energy to keep you alive.
What supplies do they need?

In order to live, cells require a source of **glucose**, **oxygen**, and **water** to efficiently convert the chemical energy stored in glucose into ATP*.

Of course, there are more “things” involved, but in this part we are going to focus mainly on these three.

*ATP – your body’s special energy carrier – a molecule made of ten carbon atoms, sixteen hydrogen atoms, five nitrogen atoms, thirteen oxygen atoms, and three phosphorus atoms.

first thing (it’s hard to tell which one is first. For the need of not making this any harder, lets conclude that you eat first) you eat. And this is what is happening with a glucose molecule (occuring in foods) in your body



useful fact about water part in your energy maintainance: when you don’t drink enough water your blood plasma can become thicker and your heart and veins cells may need more energy to pump it. Therefore you have less energy bcause you need to produce more and pump harder

1

Glucose is gradually broken down into smaller molecules through a series of chemical reactions taking place in your cells.

2

The electrons released during glucose breakdown (after breaking, glucose changes into molecules that can only maintain smaller amounts of electrons) are transferred onto the electron transport chain in the inner *mitochondrial* membrane* from the inside of the *mitochondrion, called the matrix.*

3

**A mitochondrion is an organelle inside a cell where the parts of glucose processing and ATP production is taking place.*

The movement of electrons through protein structures in the inner mitochondrial membrane provides energy for these structures to pump protons* out of the matrix, which is inside the mitochondrion, and into the intermembrane space.

4

Then, because of the increasing amount of protons in the intermembrane space, some of them 'naturally'/'with no force needed' go back into the matrix by using another fiber-based structure. Because they moved through a fibre, they left some energy to it, that provides creating more ATP.

5

6

**protons are hydrogen atoms, without the electron. So it is just proton and neutron with the possibility of becoming a hydrogen (if added electron).*

At 'the end' of the electron transport chain, oxygen gains 2 extra electrons from it.

7

And now, two protons and one oxygen with two extra electrons, once connected, create a water molecule.

Precisely, it happens with two oxygens, O_2, which together obtain four extra electrons from the membrane, and four protons - four hydrogens, each with no electrons - so eventually two water molecules are created.

Both water and ATP leave mitochondrion matrix into the outer mitochondrion, inner cell space.

8

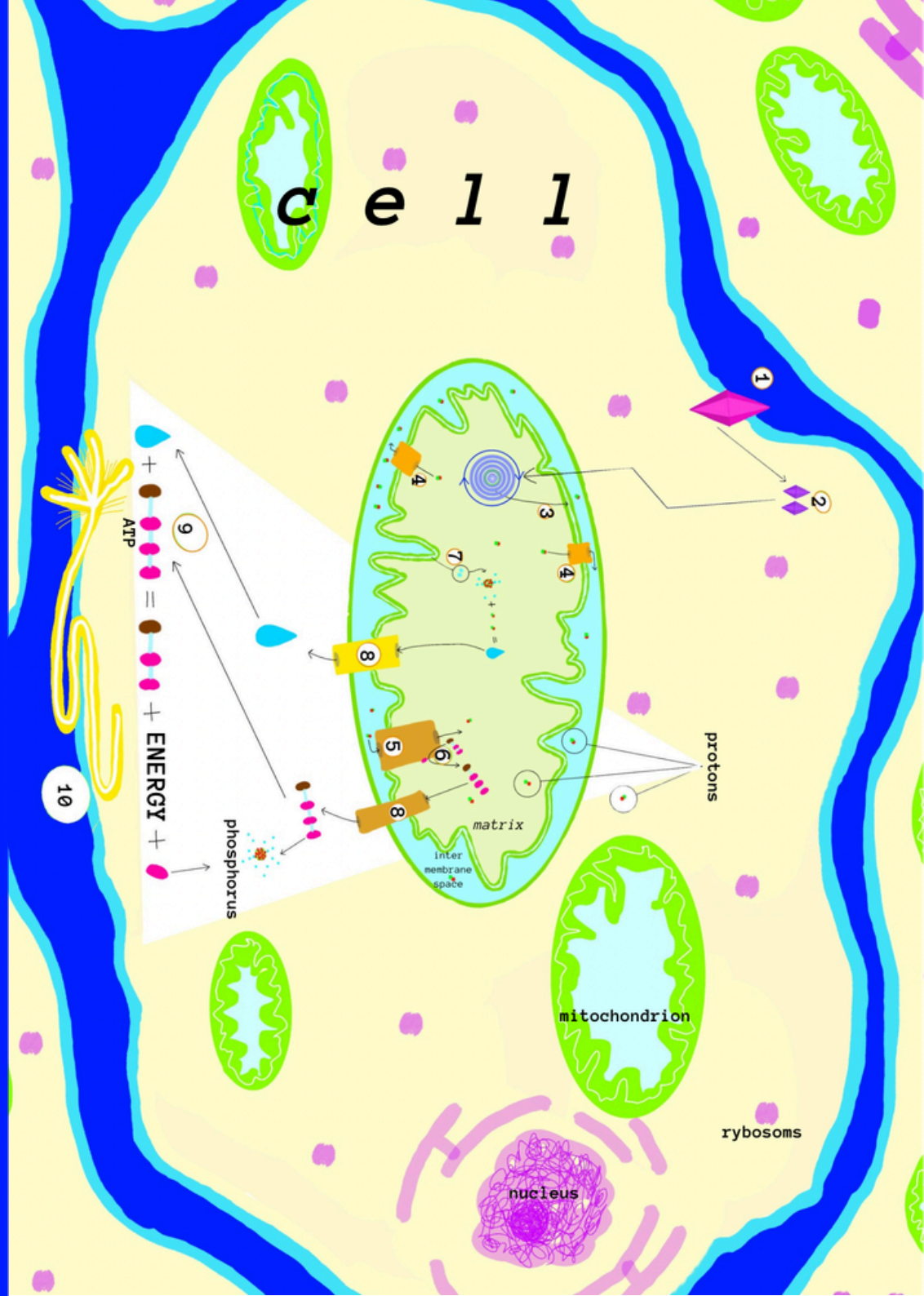
Water helps to obtain energy from one of ATP's phosphate bonds.

9

Once one of the bonds is broken, the energy used to maintain the bond is being released as heat that your body can use.

10

Singular cell can provide from 10 mil to 10 mld of ATP per one second.



So you need oxygen, water and source of glucose

ANOTHER TOPIC

Everything wrong with our crucial worlds matter maintainance

The basic needs for humans to live are:
water
oxygen
source of glucose (food)

All of these are available on Earth in shared, long lasting sources.

All sources of water and oxygen are from the same 'source' for EVERYONE. Everyone just obtains water and air from different areas of the same source.

Each countries water and air maintainance directly influences the other countries water and air conditions.
Mainly because of the forces we cannot control, that are dependent on Earth position in relation to sun and moon (currents) and differences in local temperature (winds).
So if one country pollutes their rivers, the pollution will get to the ocean and spread worldwide, or if a country burns plastic, the chemical fog will travel via winds, to other parts of the world.

We mainly obtain water from underground sources. Which is very harmfull and depleating for local fauna and flora.

there are other ways to obtain safe water. Some countries do this by collecting rain water, cleaning the already used water and by seawater desalination.

But what will happen if we continue to overuse underground waters (as we do now) is:

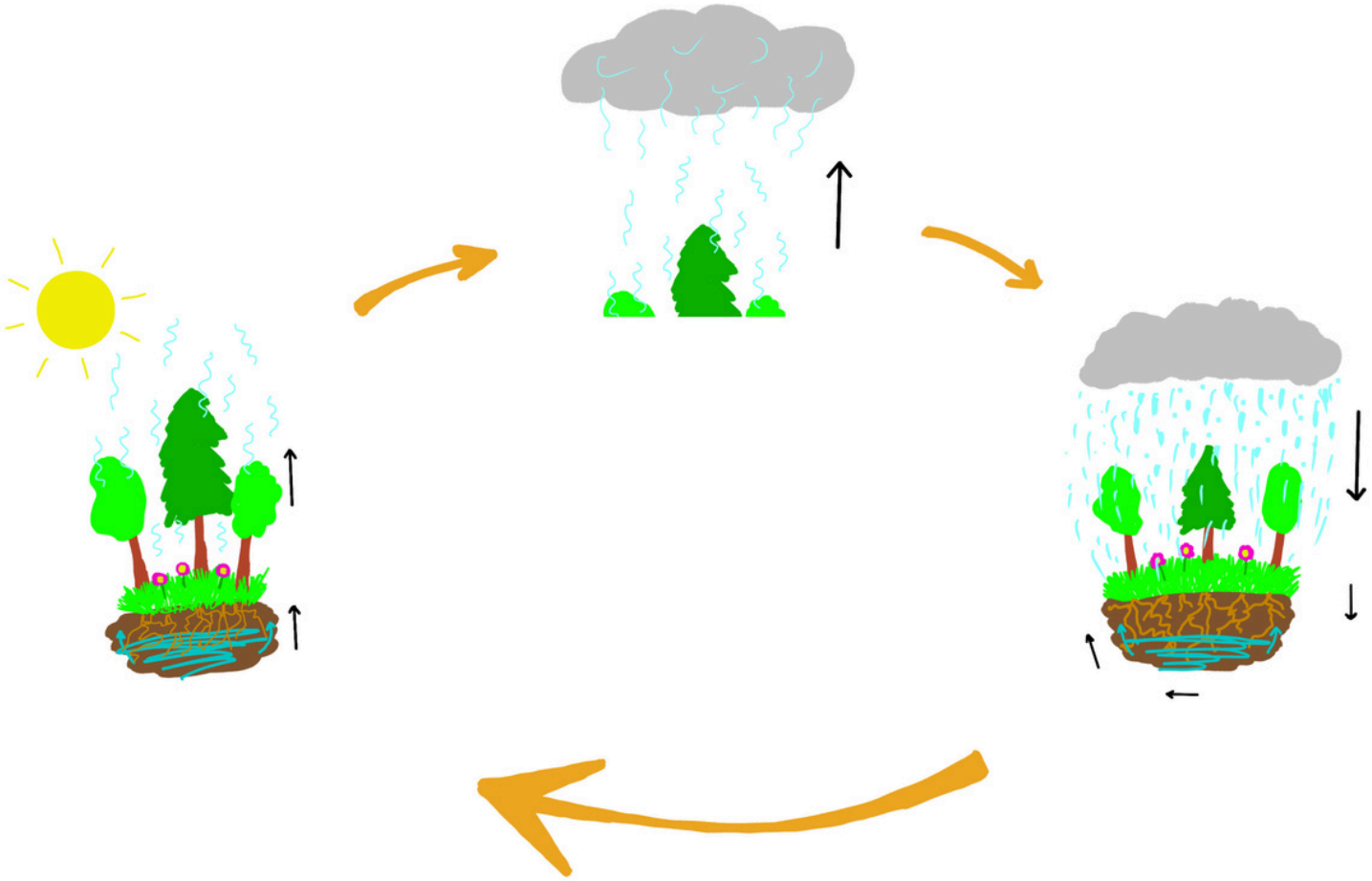
decreasing the levels of underground water, making it harder and harder and finally impossible for local flora to obtain it (because flowers obtain water mainly from roots).

And when big local areas seemingly decreases their levels of vegetations, natural cycles (including rains) can stop there.

And preserving local vegetation is the key to preserving the natural and endless cycles of water.

plants absorb water from the soil through roots.
The absorbed water evaporates to the atmosphere, through leaves.
Water vapor contributes to the formation of rain clouds.
Rain clouds contribute to rain.
Rain rehydrates the soil and lets the cycle begin again.

Therefore plants act like a natural pump.



Excessive dying off vegetation on Earth can cause dramatic and irreversible changes for a long time (droughts and floods). So it is very important for us (creatures 100 procent dependent on water) to care about its presence.

The way we obtain water is not the only issue for us.

We obtain water from the underground because ground waters are polluted.

And we dispose water to everyone in plastic, because plastic is cheap.

Brief history of how water was distributed:

3000 B.C.E – 1920 A.D
Water was transported in lead pipes into ceramic glass or vessels.
It lasted until science got developed enough to state without a doubt, that water in which lead is present has harmful effects on health.

1920 (and earlier) A.D – 1980 A.D
Water was transported in stainless steel pipes into glass bottles.
It lasted until cheaper and lighter material – plastic – was introduced.

1980 A.D – now
Water is transported in plastic pipes into plastic bottles.

In EU plastic bottles and containers for water, food, cosmetics etc. Are allowed to release into the product up to 10mg of micropastics* for 1dm2. In other parts of the world the regulations on that matter are less clear.

Plastic became widely used in 1930, and before this time our cells have never seen anything similar to plastic, so they poorly manage to digest and detoxicate it. Therefore plastic particles can be stored in your body.

Plastic on its molecular level is porous and nothing can change that. So the quality of the plastic doesn't matter, there still will be the migration of particles

*micro and nano plastics are teeny, tiny plastic parts

Since plastic is widely used for around 4five years (now, when its 2026), the studies on long term, direct plastic exposure are still not yet fully developed to determine all health effects unequivocally.

HOWEVER

There are already studies and growing evidence that micro- and nanoplastic, as well as compounds that may migrate from plastic materials into water, food and cosmetics, may not have a good effect on health. There are indicators, that among other things, can disrupt hormonal balance.

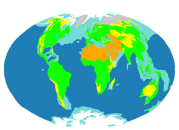
There is one widely known study provided by the Robert Koch Institute (RKI) and the German Environment Agency about how much microplastic, unable to leave our bodies, we have in our blood, pee, and lungs. From a study on 2,500 children and teenagers, it turned out that 97% of them had some amount of plastic in them. However, the symptoms of long-term exposure are, of course, still unclear...

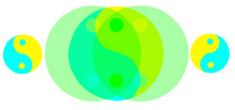
From what we can speculate, the effects of long plastic exposure may look like:

– **chronic inflammation**, which may look like: constant lack of energy, trouble sleeping, unexplained muscle pain, acne. You can support your cells by not giving them extra toxins to digest: less sugar, less processed food, less unhealthy fats, and some extra fruits and vegetables of course.

– **oxidative stress**, which means that your cells can't produce enough healthy cells that could fight toxins. This may look like: chronic fatigue, headaches, concentration problems, dry skin, and weakened immunity. You can support your cells by giving them the right fuel, in this case food rich in vitamins A, C, and B: carrots, sweet potatoes, spinach, mango, lemon, kiwi, broccoli, avocado, olives, almonds.

– **the hormonal system**. In this case, a little understanding of **how the hormonal system works** may be helpful: hormones are little fibers that connect to specific receptors in your body. Once the receptor receives a specific hormone, some body activities like, for instance, producing immune cells, lowering blood pressure, or maintaining sugar levels can start. And what micro- and nanoplastics do is clog these receptors, so certain natural reactions can be limited. An unbalanced hormonal system may look like: fertility disorders, chronic fatigue, sudden weight loss or gain, low sex drive, acne, annoying mood swings. You can help your cells produce hormones and unclog receptors by: having lots of good quality sleep, working out, and eating lots of fish (e.g. salmon), eggs, fruit, chia and pumpkin seeds, oysters, spinach.





So what is now:

We obtain water from underground sources, which influences the environment dramatically and is heading towards ending crucial for every living creature on Earth natural water cycles.

We consume water that is given to us in plastic bottles via plastic pipes, which directly and badly influences our health.

some help for yourself:

As individuals, we can limit our exposure to plastic by using reusable cups. They reduce the time water spends in plastic (the longer water is in plastic, the more of it it absorbs) and this also reduces the use of disposable plastic cups (why this is also crucial is described later in the text). Also not reheating your foods in plastic containers. It altogether amounts to a smaller quantity of microplastics in the body and a potential improvement in various health aspects.

It's okay, that some countries dispose water globally, but it's not okay how they obtain and dispose it. Especially when we already know, that the way they are doing that has a destructive impact both on Peoples and Earth condition.

The solution may not be about corrupting the already existing huge companies, that maintain the waters and plastics, but it can be about relocating their finances.

for instance →

Invest in plastic polution maintanance = clean ground waters.

Sell picked up plastic to companies, that makes plastic reusable.

Sell mines into their parts firsts (and obtain stainless steel).

With money from collected plastic and old mines parts, build infrastructures that can obtain water from both rain and ground waters (since now after you have a sewage treatments these are also clean water sources).

Save money, because rain collecting and filtration systems are cheaper to maintain, plus if your country has rain you can be less dependent on imported water.
And since we have a free-market economy, companies that have groundwater mines can reinvest their capital into building infrastructure in equatorial countries, where rainwater is omnipresent (of course with appropriate regulations regarding how this insfrastucture may work, so that it does not destroy local people, fauna and flora lifes).

There are countries from whoose solutions we can learn

Singapoire

A country in which every citizen has access to drinkable water

they achieced that by:

cleaning and monitoring rivers, and collecting rainwater using special tanks

remaining issue

water is still being mainly distributed via plastic pipes and bottles

Netherlands

A country that leads the way in plastic waste management. It implements projects such as the construction of roads, benches, and bridges using plastic

they achieced that by:

river cleanup and monitoring, waste recycling, and the development of new infrastructure projects using reclaimed waste as building material

remaining issue

the same as Singapoire

Poland

collecting plastic water bottles and cans and disposing them into special bins (occuring for example in shops), allows people who collect them to make some extra money. Which certainly helps people in poverty crisis.

ADDITIONALLY

All states have adapted their economies to global solutions, such as contemporary cars. Some states specialize in the production of certain car parts, some in assembly, some in electronics and some in sales. Thanks to cooperation on a shared, globally used idea for transport, every country and company has the option to take up a part in this global idea realization and gain some money from it.
Cars as we know today are globally the same.
They could have looked and worked differently, but there is no need for that, since they work well (especially with electric vehicles, which do not emit CO2 during driving).

So maybe there is a chance to find a global solution to maintain water and pollution and work on it globally (Aspecially since it is a topic that undeniably concerns EVERYONE)?
And maybe there is a chance to change how we treat and dispose water globally?
Since every country has adapted to the plastic solution, it can adapt to any other. Aspecially since we are humans - creatures who can adapt and evolve easily thanks to their brains.

We changed the water infrastructure which existed for 5000 years, the moment we realised it was harmful.
Now, similarly we start to confirm with certainty, that our latest solutions are harmful too. So there just is a need to change.

If we can take a clear pictures of Saturn and build quantum computers, we surely have all what we need to obtain and dispose water logically.

And thanks to history we learned, that stainless steel pipes and glass bottles were the safest. So there may not be the need to investigate, but to act on what we know.

And is is like that? - if we know facts, and so we know that plastic based water infrastructure and underground water deplation is bad for human health and entire planet state, then if one country doesn't want to change that, wouldn't it means it delivers solutions that are openly working towards everyone's harm.

*How we can use the recycled plastic *IDEAS*

Collected and recycled plastic from plastic pollution, plastic bottles and plastic pipes can be reused for example to make easy to use and appealing water obtaining points and garbage bins in public spaces, shops etc.



It can be also used to create multiuse bottles or cups available next to the 'water wending machines' for those who don't have or have forgotten their multiuse cup.
It's still plastic, but it's denser and water is not in it for a long time, like it is in regular shops in regular plastic bottles. Plus it's easier to recycle since we can also build different purpose plastic bins out of the recycled plastic. For example the ones for the reusable plastic cups, we can put next to the water obtaining points.

in Thailand there are water collecting points, that look like this

Their desing has some space for improvement, however they work in a very beautifull way. You put any amount of money and you pour as much water s you need.

And isn't it like that? – if we cooperate, there is a bigger chance, that making a global change concerning topics that affects EVERYONE, will be both easier (smoother, with less misunderstandings) and faster?

*How we can use the recycled plastic *IDEAS*

Different looking bins for different trash (plastic, paper, bio, glass etc.).

There is no restriction on how can they look and there are no regulations about how many can be there. So why not creating them in a way everyone will understand their purpose and will want too use them and can asily access them?

Paper

Buisnes related to making paper out of plastic sounds like a brilliant environmental and money wise plan.
Imagine if we switched from tree based paper to plastic waste based paper.

less trees would need to be cut (trees crucial for oxygen production and rain circulation)
And more plastic can be collected and recycled to obtain the material (contributing to create cleaner environment)

Additionally plastic cn be way more durable than regular paper.

There is even a chance it can be done without the need to melt the plastic (and therefore without creating extra pollution in the atmosphere).
Shredding the plastic, mixing it with the right chemicals, forming, pressing and drying it. Sounds like a sensible and doable idea.
And for those who will experience the change (for example books users) it would be just about getting used to new structure of ‘paper’. What does it matter if it is white with black printed text on it or if it is a little colorful and probably a little grey with black printed text on it.

Benches, bridges, water obtaining points, tables, furniture,

There is no limit on shapes, structures colours and sizes, since plastic is an easy to shape material

Also since we can almost certainly tell that plastic occuring in our bodies is harmful for them, why would we use plastic for clothes (clothes produces microplastic dust, which we inhale) or to keep our foods and cosmetics in it?

There are so many better ways. Change is not easy. But once it’s done you suddenly realise you live in a sustainable, clean and fun to live in world.

BRIEF SUMMARY IN A FORM OF A ALMOST PERFECT WORLD VISION

Every Earth inhabitant has the access to clean and drinkable water source, from global water solution infrastructure.

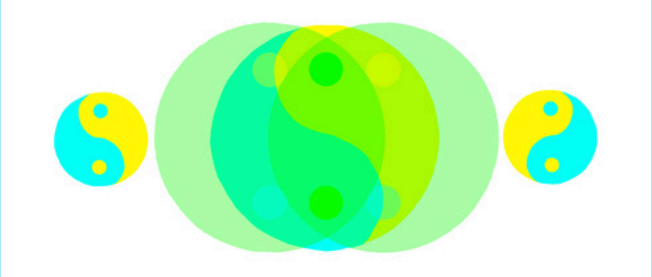
Water is delivered through stainless steel pipes and in glass bottles or aluminum cans or maybe even paper cartoons (because why not, since juice companies sells their juice this way). Globally it travels in big stainless steel containers and it is later divided into for example glass bottles, so the water loss due to transportation decreases.

You can obtain safe water from your home or from globally and easily accessible water points. You can obtain as much as you want for a certain and specified fee. You can either buy a water container in a vending machine or you can bring yours.
Such points you can find in public spaces, shops and in neighbourhood areas.
And in a perfect vision they look and work the same world wide, so there is no worry about having access to safe and drinkable water anywhere.

There is hardly any trash on the streets and in waters, since water and street purification was and is needed and mandatory for every state to create and maintain safe water sources and new usable materials sources.
You can also easily maintain your own trash, because the small size and general lack of bins in not an issue anymore.

There is a new global organization which creates global ideas and solutions fully and almost only depending on knowledge. It only creates soultions that are supporting life and concern global topics.
Countries representatives and individuals can vote on different ideas and solutions, which need to be considered similarly globally. This is crucial, because as written earlier if one country treats their waters and air insufficiently it directly influences countries that treat the crucial sources sufficiently.

Nature and Humanity is thriving. Crucial cycles are preserved.



sources and links

Robert Koch studies

https://www.greenqueen.com.hk/microplastic-pollution-german-study-finds-plastic-byproducts-in-over-97-of-children-tested/

Singapoire water menagement

https://www.pub.gov.sg/Public/WaterLoop/OurWaterStory

Netherlands on plastic solutions

https://materialdistrict.com/article/how-a-dutch-factory-is-turning-plastic-waste-into-sustainable-panels-and-boards/

on water in general

https://en.wikipedia.org/wiki/Water

Every knowledge introduced in this magazine was provided by studies, lectures and books. Entire magazine is written by human with a support of ai tools for detecting possible scientific mistakes.

Thanks for reading. New conclusions soon.

in the next magazine:

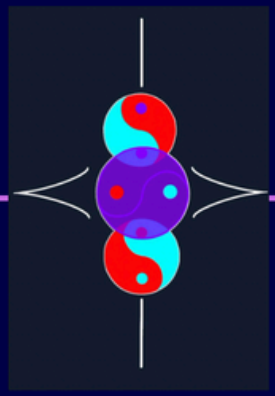
first part on light and sound. Why is it more than you can 'see'/'hear', why does it matter and why is it crucial

what your brain likes? How to not act only on it and still let it thrive

what was and still can be smarter, more available, less annoying and more useful than a smart phone?

popular theories on what happens after you die. Analysis

and more ideas on how can the World be improven



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